

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B5: GENERAL RELATIVITY

TRINITY TERM 2021

Thursday, 17 June

Opening time: 9.30 am UK Time

**You have two hours writing time to complete the paper
and up to 30 minutes technical time to upload your answer file.**

*Answer **two** questions.*

Write your candidate number, paper name, paper code and a list of the questions that you have answered on the first page, and submit all of your answers as a single PDF file.

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors will be available within Inspira during the examination

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

Expressed in terms of the Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$, the Riemann tensor has components

$$R^\gamma_{\alpha\beta\delta} = \frac{\partial}{\partial x^\delta} \Gamma_{\alpha\beta}^\gamma - \frac{\partial}{\partial x^\beta} \Gamma_{\alpha\delta}^\gamma + \Gamma_{\alpha\beta}^\epsilon \Gamma_{\delta\epsilon}^\gamma - \Gamma_{\alpha\delta}^\epsilon \Gamma_{\beta\epsilon}^\gamma.$$

The Ricci tensor is defined as $R_{\alpha\beta} \equiv R^\gamma_{\alpha\beta\gamma}$. When expressed in a basis in which the metric tensor $g_{\alpha\beta}$ is diagonal, it can be written as

$$R_{\alpha\beta} = \frac{1}{2} \frac{\partial^2}{\partial x^\alpha \partial x^\beta} \ln |g| - \frac{\partial \Gamma_{\alpha\beta}^\gamma}{\partial x^\gamma} + \Gamma_{\alpha\gamma}^\delta \Gamma_{\beta\delta}^\gamma - \frac{1}{2} \Gamma_{\alpha\beta}^\gamma \frac{\partial}{\partial x^\gamma} \ln |g|,$$

where $|g|$ is the modulus of the determinant of $g_{\alpha\beta}$.

The Einstein equation is

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R + \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta},$$

where $R \equiv g^{\alpha\beta} R_{\alpha\beta}$ is the Ricci scalar, Λ is the cosmological constant and $T_{\alpha\beta}$ is the energy-momentum tensor.

For the line element

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right]$$

the Einstein equation can be written as the pair of equations

$$\begin{aligned} \ddot{R} &= -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) R + \frac{1}{3} \Lambda c^2 R, \\ \dot{R}^2 &= \frac{8\pi G}{3} \rho R^2 + \frac{1}{3} \Lambda c^2 R^2 - c^2 k, \end{aligned}$$

where ρ and p are the rest-frame density and pressure, respectively.

1. In a world where General Relativity is the only known theory of gravity, consider why an apple falls from a height h on the tree to the ground. Start your consideration from the moment the stem of the apple breaks completely, and the apple is, for a moment, free to embark on its natural path (it is fine to assume the apple as a point particle and its surrounding medium to be vacuum). Before you begin and to get you in the right mindset, write down one sentence that describes gravity in the context of General Relativity. [2]

(a) Start by the equation of motion in the apple's own local inertial coordinates $d^2\xi/d\tau^2 = 0$. Define τ and derive the geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} + c^2\Gamma_{00}^\mu \left(\frac{dt}{d\tau}\right)^2 = 0$$

and define its parameters. [4]

(b) Choose the known metric $g_{\mu\nu}$ from the appropriate solution of Einstein's field equations to derive the radial motion of the apple towards the ground as seen by the observer. Write down and explain your choice of metric, explain the terms, and justify the approximations you are making. [5]

(c) For the geodesic equation in (a) evaluate the relevant affine connections, and show explicitly that the motion will be in the radial direction. Derive the form of r versus time measured by a nearby observer who is sitting under the tree. Again, explain your approximations, particularly your choice of time. [7]

(d) Assume the apple has unit mass, and use this and your chosen metric to establish the constant total energy of the apple at the moment the stem snaps. Show that the gain in velocity when the apple reaches the ground obeys

$$\frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 = GM \left(\frac{1}{R_E} - \frac{1}{R_E + h}\right),$$

where R_E is the radius M is the mass of the Earth, and G the gravitational constant. [7]

2. (a) Which of the following systems will emit classical gravitational waves? Explain your answers.

- (i) A motionless point particle.
- (ii) A rotating sphere of fixed centre of mass and constant density.
- (iii) A disc rotating around an axis perpendicular to it and passing through its centre.
- (iv) A disc, with its normal axis making an angle $\alpha \neq 0$ relative to the vertical z -axis, with the disc precessing around the z -axis at some rate ω .

[8]

(b) A gravitational wave source consisting of two point mass spheres (each of mass M) connected by a rigid massless rod of length $2l$ is rotating at angular velocity ω in the xy plane. A long distance r_0 away from the source, there are four free masses (each of mass m) directly above the source. At $t = 0$, two of the four free masses are located at $x = \pm a, y = z = 0$ and two at $x = z = 0, y = \pm a$. From the point-of-view of the free masses, the source lies directly overhead along the z axis. Assuming the source has been rotating for a very long time, what are the metric components $h_{xx}, h_{yy}, h_{xy}, h_{yx}$ that are generated by the rotating masses? These should be expressed as a function of z and time t . Define any other symbols you use.

[5]

(c) What are the displacements ξ_x, ξ_y, ξ_z of each of the free masses m as the gravitational wave passes? Sketch the displacements over one full wave cycle assuming the initial placement described in part (b). Does the motion correspond to the $+$ mode, the $\times$ mode, or a superposition of the two?

[4]

(d) We have often remarked that gravitational radiation (GR) is analogous to electromagnetic (EM) radiation. EM can exhibit (i) polarisation, (ii) interference, (iii) refraction (with an index of refraction n), and (iv) diffraction. Discuss both the possibility and also the plausibility of applying each of these concepts to gravitational radiation. A few sentences for each of these should be sufficient.

[8]

3. (a) Consider a Schwarzschild black hole. Show that the rate of passage of time t for any circular orbit is given by:

$$u^0 = \frac{dt}{d\tau} = \left(1 - \frac{3GM}{rc^2}\right)^{-1/2},$$

and explain all the terms (u^0, t, τ, G, M, r, c).

[5]

(b) Consider a space-like 4-vector with no 0-component in the rest frame, for example, the spin $\mathbf{s}$ of a body in such a circular orbit of radius r , at some time $t = 0$. Assume $\mathbf{s}$ is pointing along the radial direction at $t = 0$. Following the equation of parallel transport, show that the orientation of $\mathbf{s}$ will rotate with respect to the radial direction at an angular rate of $\Omega_r = \Omega/u^0$, where $\Omega = d\phi/dt$ and ϕ is the coordinate angle along the orbital plane. Derive the equations showing how the components of $\mathbf{s}$ change with coordinate time. Use r, θ, ϕ coordinates.

[8]

(c) The spaceship Rocinante has positioned itself in a circular orbit around a non-spinning black hole of mass $M = 8.2 \times 10^{36}$ kg, at some coordinate distance $R = 200R_s$, where R_s is the Schwarzschild radius. Assume that at great distance, Earth is also on the same plane as the orbit of the Rocinante. On board, a freely spinning gyroscope is used to direct a narrow beam of light for communications purposes. There are no frictional losses.

Work out the orbital periods of the Rocinante, as observed by a distant observer, and as computed by the crew on board. Explain the difference.

[4]

(d) The narrow communications beam has a circular cross-section and an angular diameter of 1 degree. Assuming the beam was pointed exactly along the radial direction towards Earth at t_0 , will the beam be pointing towards Earth after exactly 1 orbit? Is distortion of the beam path due to the local spacetime curvature important?

[8]

4. (a) Write down the Friedmann-Robertson-Walker metric that best describes our Universe. How does this differ from the most general FRW metric? Be precise and define all the terms that you use. [5]

(b) Give a brief explanation of how equation:

$$\dot{\rho} + 3(\rho + P/c^2)\frac{\dot{R}}{R} = 0$$

emerges in cosmological GR. Define the terms and prove that it is equivalent to the first law of thermodynamics, $dE = -PdV$, where E is the energy within a comoving volume V and P is the pressure. [5]

(c) Consider a possible history of a flat Universe with 3 phases characterised by 3 equations of state: S1 when $P = -\rho c^2$ initially, then S2 when $P = \frac{1}{3}\rho c^2$, and finally phase S3 when $P = 0$. What is the physical interpretation of each of these three equations of state? [6]

(d) Write down an expression for the age of the universe in terms of an integral over the Hubble parameter. For the model of the Universe described in (c) and assuming the initial size of the Universe to tend to 0, show that a Universe including S1 must be older than a Universe without S1. [5]

(e) Discuss how the real Universe differs from the model proposed in part (c). [4]